Supervised Bayesian Matrix Factorization

Corrected Update Derivations with Full Step-by-Step Algebra

Revision of the February 12, 2026 Document

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March 2026

Contents

1 Introduction

This document presents the full derivations for the Supervised Bayesian Matrix Factorization (Supervised MF) model. It covers the objective function (ELBO), the second-order Taylor approximation of the Cox survival likelihood, and the complete coordinate-ascent updates for the variational posteriors $q(l_k)$, $q(f_k)$, $q(\beta_k)$, and the noise precision τ.

Every algebraic step is shown explicitly. Correction boxes flag the eight errors found in the February 12, 2026 document (R1–R8).

1.1 Update Procedure

The general procedure for prior and posterior updates is:

For $b = 1, \ldots, B$:

1. Compute the Cox Taylor working quantities z_i, W_{ii}.
2. Iterate over $k = 1, \ldots, K$ for each of the following:
   (a) Update $q(l_k)$, $g(l_k)$ with all other parameters fixed (combines genomics and survival terms — differs from standard EBMF).
   (b) Update $q(f_k)$, $g(f_k)$ with all other parameters fixed (identical to the standard EBMF problem).
   (c) Update $q(\beta_k)$, $g(\beta_k)$ with all other parameters fixed (survival term only — differs from standard EBMF).
3. Update τ_{ij} (identical to the standard EBMF problem).

2 Notation and Model Setup

2.1 Index Conventions

Symbol	Dimension	Meaning
n	scalar	number of samples (patients)
p	scalar	number of features (genes / CpG sites)
K	scalar	number of latent factors
i	$1 \ldots n$	sample index
j	$1 \ldots p$	feature index
k, k'	$1 \ldots K$	factor index
Y	$n \times p$	observed genomics data matrix
L	$n \times K$	loading matrix; l_{ik} = loading of sample i on factor k
F	$p \times K$	factor matrix; f_{jk} = weight of feature j on factor k
$\boldsymbol{\beta}$	$K \times 1$	survival coefficient vector
τ	$p \times 1$	column-specific noise precision (inverse noise variance)
(t_i, δ_i)		survival time and event indicator for sample i

2.2 Posterior Moment Notation

Throughout, $q(\cdot)$ denotes the variational posterior and $g(\cdot)$ the prior. We carefully distinguish the *posterior mean* from the *posterior second moment*:

$$\bar{l}_{ik} := \mathbb{E}_q[l_{ik}] \qquad \text{(posterior mean)} \tag{1}$$

$$\overline{l^2_{ik}} := \mathbb{E}_q[l^2_{ik}] = \mathrm{Var}_q(l_{ik}) + \bar{l}^2_{ik} \qquad \text{(posterior second moment)} \tag{2}$$

and analogously $\bar{f}_{jk}$, $\overline{f^2_{jk}}$, $\bar{\beta}_k$, $\overline{\beta^2_k}$. These two quantities are **not** equal whenever there is posterior uncertainty. This distinction is essential for the precision update (Section 8) and for the error-in-variables correction in the β update (Section 7).

2.3 Models

Revision Note

[R1] The 2/12/26 document wrote $Y = L_{n\times j}F'_{n\times k} + E_{k\times j}$, which has inconsistent and impossible dimensions. Corrected below.

Genomics model.

$$Y_{n\times p} = L_{n\times K}\, F^{\top}_{K\times p} + E_{n\times p}, \qquad E_{ij} \overset{\text{iid}}{\sim} \mathcal{N}\Big(0,\, \tau_j^{-1}\Big) \tag{3}$$

Element-wise: $Y_{ij} = \sum_{k=1}^{K} l_{ik} f_{jk} + \varepsilon_{ij}$.

Survival model (Cox proportional hazards).

$$h(t_i) = h_0(t_i)\, \exp(\eta_i)\,, \qquad \eta_i = \boldsymbol{l}_i^{\top}\boldsymbol{\beta} = \sum_{k=1}^{K} l_{ik}\beta_k \tag{4}$$

where h_0 is an unspecified baseline hazard.

Hierarchical priors.

$$l_{ik} \overset{\text{iid}}{\sim} g_l \in \mathcal{G}_l, \quad f_{jk} \overset{\text{iid}}{\sim} g_f \in \mathcal{G}_f, \quad \beta_k \overset{\text{iid}}{\sim} g_\beta \in \mathcal{G}_\beta$$

All three priors are learned from data via the Empirical Bayes Normal Means (EBNM) framework.

3 Objective Function (ELBO)

The Evidence Lower Bound (ELBO) to be *maximized* over posteriors (q_l, q_f, q_β), priors (g_l, g_f, g_β), and precision τ is

$$\begin{aligned}\mathcal{F} &= \mathbb{E}_{q_l,q_f,q_\beta}[\log P(Y,t,\delta \mid L,F,\boldsymbol{\beta},\tau)] \\ &\quad + \sum_k \mathbb{E}_{q_{l_k}}\left[\log\frac{g_{l_k}(l_k)}{q_{l_k}(l_k)}\right] + \sum_k \mathbb{E}_{q_{f_k}}\left[\log\frac{g_{f_k}(f_k)}{q_{f_k}(f_k)}\right] + \sum_k \mathbb{E}_{q_{\beta_k}}\left[\log\frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)}\right]\end{aligned} \tag{5}$$

The joint log-likelihood decomposes as (conditional independence of Y and (t,δ) given L):

$$\log P(Y,t,\delta \mid L,F,\boldsymbol{\beta},\tau) = \underbrace{\log P(Y \mid L,F,\tau)}_{\text{genomics}} + \underbrace{\log P(t,\delta \mid L,\boldsymbol{\beta})}_{\text{survival}} \tag{6}$$

4 Taylor Approximation of the Cox Partial Likelihood

Because $\log P(t, \delta \mid L, \boldsymbol{\beta})$ is non-conjugate in L and $\boldsymbol{\beta}$, we use a second-order Taylor expansion to obtain a locally Gaussian surrogate that can be handled by EBNM.

4.1 The General Second-Order Taylor Expansion

The second-order Taylor expansion of a function $f(x)$ around a point x_0 is:

$$f(x) \;=\; f(x_0) \;+\; \underbrace{f'(x_0)}_{\text{gradient/score}} \;(x - x_0) \;+\; \frac{1}{2}\; \underbrace{f''(x_0)}_{\text{Hessian/info}} \;(x - x_0)^2 \tag{7}$$

4.2 Applying the Expansion to the Cox Log-Likelihood

Let $\ell(\boldsymbol{\eta}) = \log P(t, \delta \mid L, \boldsymbol{\beta})$ be the Cox partial log-likelihood, where the linear predictor for sample i is:

$$\eta_i \;=\; \boldsymbol{l}_i^\top \boldsymbol{\beta} \;=\; \sum_k l_{ik} \beta_k \tag{8}$$

Let $\hat{\boldsymbol{\eta}}$ denote the current expansion point. Define the score vector $\mathbf{u}$ and the negative diagonal Hessian W as:

$$u_i = \left.\frac{\partial \ell}{\partial \eta_i}\right|_{\hat{\boldsymbol{\eta}}} \quad \text{(Cox score for sample } i\text{)} \tag{9}$$

$$W_{ii} = -\left.\frac{\partial^2 \ell}{\partial \eta_i^2}\right|_{\hat{\boldsymbol{\eta}}} \;\geq\; 0 \quad \text{(negative diagonal Hessian, always non-negative)} \tag{10}$$

Applying the second-order expansion (Eq. (7)) to $\ell(\boldsymbol{\eta})$ around $\hat{\boldsymbol{\eta}}$:

Derivation Steps

$$\ell(\boldsymbol{\eta}) \approx \ell(\hat{\boldsymbol{\eta}}) + (\boldsymbol{\eta} - \hat{\boldsymbol{\eta}})^\top \underbrace{\nabla \ell(\hat{\boldsymbol{\eta}})}_{\mathbf{u}} + \frac{1}{2}(\boldsymbol{\eta} - \hat{\boldsymbol{\eta}})^\top \underbrace{\nabla^2 \ell(\hat{\boldsymbol{\eta}})}_{H=-W}(\boldsymbol{\eta} - \hat{\boldsymbol{\eta}}) \tag{11}$$

$$= \ell(\hat{\boldsymbol{\eta}}) + \sum_i (\eta_i - \hat{\eta}_i)\, u_i + \frac{1}{2} \sum_i (\eta_i - \hat{\eta}_i)^2\, H_{ii} \tag{12}$$

Since $H_{ii} = -W_{ii}$ (negative Hessian), substitute:

$$= \ell(\hat{\boldsymbol{\eta}}) + \sum_i (\eta_i - \hat{\eta}_i)\, u_i - \frac{1}{2} \sum_i (\eta_i - \hat{\eta}_i)^2\, W_{ii} \tag{13}$$

Expand $(\eta_i - \hat{\eta}_i)^2 = \eta_i^2 - 2\eta_i\hat{\eta}_i + \hat{\eta}_i^2$:

$$= \sum_i \left[u_i(\eta_i - \hat{\eta}_i) - \frac{1}{2} W_{ii} \left(\eta_i^2 - 2\eta_i\hat{\eta}_i + \hat{\eta}_i^2 \right) \right] + C \tag{14}$$

Distribute and retain only terms involving η_i (constants w.r.t. η_i absorbed into C):

$$= \sum_i \left[u_i\eta_i - \frac{1}{2} W_{ii}\eta_i^2 + W_{ii}\hat{\eta}_i\eta_i \right] + C \tag{15}$$

Group into the quadratic form $\sum_i \big(a_i\eta_i^2 + b_i\eta_i \big)$, where:

$$a_i = -\tfrac{1}{2} W_{ii}, \qquad b_i = u_i + W_{ii}\hat{\eta}_i \tag{16}$$

Completing the square: For any quadratic $a_i\eta_i^2 + b_i\eta_i$ with $a_i < 0$:

$$\sum_i \big(a_i\eta_i^2 + b_i\eta_i \big) = \sum_i a_i \left(\eta_i + \frac{b_i}{2a_i} \right)^2 + C = \sum_i a_i(\eta_i - c_i)^2 + C \tag{17}$$

where $c_i = -b_i/(2a_i)$. Substituting $a_i = -\frac{1}{2}W_{ii}$ and solving for c_i:

$$c_i = -\frac{b_i}{2a_i} = -\frac{u_i + W_{ii}\hat{\eta}_i}{2 \cdot (-\frac{1}{2}W_{ii})} = \frac{u_i + W_{ii}\hat{\eta}_i}{W_{ii}} = \frac{u_i}{W_{ii}} + \hat{\eta}_i \tag{18}$$

Apply the complete-the-square form back to Eq. (15):

$$= \sum_i \left(-\frac{1}{2} W_{ii} \right) (\eta_i - c_i)^2 + C \tag{19}$$

$$= \sum_i \left(-\frac{1}{2} W_{ii} \right) \big(\eta_i^2 - 2\eta_i c_i + c_i^2 \big) + C \tag{20}$$

4.3 The Working Response

Collecting results, the Cox log-likelihood is approximated as:

$$\ell(\boldsymbol{\eta}) \;\approx\; \sum_i \left[-\frac{1}{2}\,W_{ii}\,(\eta_i - z_i)^2\right] + C \tag{21}$$

where the **working response** is:

$$\boxed{z_i \;=\; \hat{\eta}_i + \frac{u_i}{W_{ii}}} \tag{22}$$

This is a weighted Gaussian likelihood with response z_i, precision W_{ii}, and mean $\eta_i = \sum_k l_{ik}\beta_k$. The quadratic surrogate is a proper likelihood for a linear model, enabling EBNM to be applied directly to both L and $\boldsymbol{\beta}$.

5 Update for q_{l_k} — Patient Loadings

5.1 Setting Up the Objective

To update $q(l_k)$ and $g(l_k)$, we extract from the full ELBO only the terms that depend on l_k (or l_k^2):

$$\mathcal{F}(q_l, g_l) = \mathbb{E}_{q_l}[\log P(Y, t, \delta \mid L, F, \boldsymbol{\beta}, \tau)] + \sum_k \mathbb{E}_{q_{l_k}}\left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)}\right] \tag{23}$$

$$= \underbrace{\log P(Y \mid L, F, \tau)}_{\text{Eq. 1: Genomics}} + \underbrace{\log P(t, \delta \mid L, \boldsymbol{\beta})}_{\text{Eq. 2: Survival}} + \underbrace{\sum_k \mathbb{E}_{q_{l_k}}\left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)}\right]}_{\text{KL terms (kept)}} \tag{24}$$

We need to identify the coefficients A_{ik} and B_{ik} attached to the terms l_{ik}^2 and l_{ik} in Equations 1 and 2, then combine them.

5.2 Defining Partial Residuals

Genomics partial residual. The full residual for entry (i, j) is:

$$R_{ij} = Y_{ij} - \sum_k l_{ik} f_{jk} \tag{25}$$

$$= Y_{ij} - \left[l_{ik} f_{jk} + \sum_{k' \neq k} l_{ik'} f_{jk'}\right] \tag{26}$$

Rearranging to isolate the k-th factor:

$$R_{ij} = \underbrace{\left(Y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'}\right)}_{R_{ij}^{-k}} - l_{ik} f_{jk} \tag{27}$$

The k**-th genomics partial residual** R_{ij}^{-k} is the data residual after removing all factors *except* k:

$$R_{ij}^{-k} = Y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'} \quad \Rightarrow \quad Y_{ij} \;\approx\; R_{ij}^{-k} + l_{ik} f_{jk} \tag{28}$$

Revision Note

[R2] The 2/12/26 document (Eq. 22) wrote $z_i^{-k} = z_i - \sum_{k' \neq k} l_{ik'} f_{jk'}$,
factor weight $f_{jk'}$ (which carries a *feature* index j) in a *per-sample*
response z_i is a survival quantity and does not involve F at all — it i
definition is:

Survival partial working response.

$$z_i^{-k} = z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'} \quad \Rightarrow \quad z_i \; \approx \; z_i^{-k} + l_{ik} \beta_k \tag{29}$$

5.3 Genomics Likelihood Contribution (Equation 1)

The genomics log-likelihood for sample i, feature j is:

$$\log P(Y_{ij} \mid l_{ik}, f_{jk}, \tau_j) = -\frac{1}{2}\tau_j \left(Y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 + C \tag{30}$$

Derivation Steps

Substituting the partial residual decomposition $Y_{ij} = R_{ij}^{-k} + l_{ik} f_{jk}$:

$$\log P(Y \mid L, F, \tau) = \sum_{ij} \left[-\frac{1}{2}\tau_j \left(R_{ij}^{-k} - l_{ik} f_{jk} \right)^2 \right] \tag{31}$$

Expand the squared term $(R_{ij}^{-k} - l_{ik} f_{jk})^2$:

$$= \sum_{ij} \left[-\frac{1}{2}\tau_j \left(\cancel{\left(R_{ij}^{-k} \right)^2} - 2\, R_{ij}^{-k}\, l_{ik}\, f_{jk} + l_{ik}^2\, f_{jk}^2 \right) \right] \tag{32}$$

The $(R_{ij}^{-k})^2$ term is a constant with respect to l_{ik} and is absorbed into C. Retaining only l_{ik}-dependent terms and taking the expectation $\mathbb{E}_{q_f}[\cdot]$ over the posterior of F:

$$\longrightarrow \sum_{ij} \left[\tau_j\, R_{ij}^{-k}\, l_{ik}\, \underbrace{\mathbb{E}_{q_f}[f_{jk}]}_{\bar{f}_{jk}} - \frac{1}{2}\tau_j\, l_{ik}^2\, \underbrace{\mathbb{E}_{q_f}[f_{jk}^2]}_{\overline{f_{jk}^2}} \right] \tag{33}$$

Collecting the coefficients of l_{ik} (term B) and l_{ik}^2 (term A) from the genomics likelihood:

Revision Note

[R3] The 2/12/26 document wrote f_{jk}^2 and β_k^2 as bare squares. These should be posterior second moments under q_f and q_β: $\mathbb{E}_{q_f}[f_{jk}^2] = \overline{f_{jk}^2}$ and $\mathbb{E}_{q_\beta}[\beta_k^2] = \overline{\beta_k^2}$. The distinction matters because $\overline{f_{jk}^2} = \mathrm{Var}_{q_f}(f_{jk}) + \bar{f}_{jk}^2 \geq \bar{f}_{jk}^2$.

$$A_{ik}^{\text{gen}} = \sum_j \tau_j \, \overline{f_{jk}^2}, \qquad B_{ik}^{\text{gen}} = \sum_j \tau_j \, R_{ij}^{-k} \, \bar{f}_{jk} \tag{34}$$

5.4 Survival Likelihood Contribution (Equation 2)

From the Taylor approximation (Eq. (21)), with $\eta_i = \sum_k l_{ik}\beta_k$:

Derivation Steps

Substitute the partial working response decomposition $z_i = z_i^{-k} + l_{ik}\beta_k$, so $\eta_i - z_i = l_{ik}\beta_k - z_i^{-k}$:

$$\log P(t, \delta \mid L, \boldsymbol{\beta}) \approx \sum_i \left[-\frac{1}{2} W_{ii} \left(z_i^{-k} - l_{ik}\beta_k \right)^2 \right] \tag{35}$$

Expand the squared term $(z_i^{-k} - l_{ik}\beta_k)^2$:

$$= \sum_i \left[-\frac{1}{2} W_{ii} \left(\cancel{\left(z_i^{-k} \right)^2} - 2\, z_i^{-k} \, l_{ik} \, \beta_k + l_{ik}^2 \, \beta_k^2 \right) \right] \tag{36}$$

The $(z_i^{-k})^2$ term is constant w.r.t. l_{ik}. Taking the expectation $\mathbb{E}_{q_\beta}[\cdot]$ over the posterior of $\boldsymbol{\beta}$:

$$\longrightarrow \sum_i \left[W_{ii} \, z_i^{-k} \, l_{ik} \, \underbrace{\mathbb{E}_{q_\beta}[\beta_k]}_{\bar{\beta}_k} - \frac{1}{2} W_{ii} \, l_{ik}^2 \, \underbrace{\mathbb{E}_{q_\beta}[\beta_k^2]}_{\overline{\beta_k^2}} \right] \tag{37}$$

Collecting coefficients of l_{ik} and l_{ik}^2 from the survival term:

$$A_{ik}^{\text{surv}} = W_{ii} \, \overline{\beta_k^2}, \qquad B_{ik}^{\text{surv}} = W_{ii} \, z_i^{-k} \, \bar{\beta}_k \tag{38}$$

5.5 Combined Coefficients and EBNM Problem

The full objective for $q(l_k)$ takes the form:

$$\mathcal{F}(q_{l_k}, g_{l_k}) = \sum_k \left[-\frac{1}{2} A_{ik} \, l_{ik}^2 + B_{ik} \, l_{ik} + \mathbb{E}_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] \right]$$

where the combined precision (quadratic coefficient) and sufficient statistic (linear coefficient) are the sums of the genomics and survival contributions:

$$\boxed{A_{ik} = \sum_j \tau_j \, \overline{f_{jk}^2} + W_{ii} \, \overline{\beta_k^2}} \tag{39}$$

$$\boxed{B_{ik} = \sum_j \tau_j \, R_{ij}^{-k} \, \bar{f}_{jk} + W_{ii} \, z_i^{-k} \, \bar{\beta}_k} \tag{40}$$

This quadratic form $-\frac{1}{2}A_{ik}\,l_{ik}^2 + B_{ik}\,l_{ik}$ is equivalent to a Gaussian likelihood $l_{ik} \mid x_i, s_i^2 \sim \mathcal{N}(x_i, s_i^2)$ with pseudo-observation $x_i = B_{ik}/A_{ik}$ and pseudo-variance $s_i^2 = 1/A_{ik}$:

EBNM Problem

For each sample i, the update for factor k solves the EBNM problem:

$$x_i = \frac{B_{ik}}{A_{ik}} = \frac{\displaystyle\sum_j \tau_j\, R_{ij}^{-k}\, \bar{f}_{jk} + W_{ii}\, z_i^{-k}\, \bar{\beta}_k}{\displaystyle\sum_j \tau_j\, \overline{f_{jk}^2} + W_{ii}\, \overline{\beta_k^2}} \tag{41}$$

$$s_i = \frac{1}{\sqrt{A_{ik}}} \tag{42}$$

$$(\hat{g}_{l_k},\, \hat{q}_{l_k}) \;=\; \mathrm{EBNM}(\mathbf{x}, \mathbf{s}) \tag{43}$$

The EBNM solver returns the fitted prior $\hat{g}_{l_k}$ and posterior moments $\bar{l}_{ik} = \mathbb{E}_{\hat{q}}[l_{ik}]$ and $\overline{l_{ik}^2} = \mathrm{Var}_{\hat{q}}(l_{ik}) + \bar{l}_{ik}^2$.

6 Update for q_{f_k} — Biological Factors

6.1 Setting Up the Objective

F appears *only* in the genomics term of the likelihood (not in the survival term), so the update is:

$$\mathcal{F}(q_{f_k}, g_{f_k}) \;\approx\; \mathbb{E}_{q_l,q_f}[\log P(Y \mid L, F, \tau)] + \mathbb{E}_{q_{f_k}}\left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)}\right] \tag{44}$$

The genomics data model is:

$$Y_{ij} \;\sim\; \mathcal{N}\!\left(\sum_k l_{ik} f_{jk},\; \tau_j^{-1}\right) \tag{45}$$

6.2 Deriving Coefficients for f_{jk}

Derivation Steps

Expand the log-likelihood:

$$\mathbb{E}_q[\log P(Y \mid L, F, \tau)] = \sum_{ij} \mathbb{E}_q\left[-\frac{1}{2}\tau_j \left(Y_{ij} - \sum_k l_{ik} f_{jk}\right)^2\right] \tag{46}$$

Use the partial residual $R_{ij}^{-k} = Y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'}$ so that $Y_{ij} = R_{ij}^{-k} + l_{ik} f_{jk}$:

$$= \sum_{ij} \mathbb{E}_q\left[-\frac{1}{2}\tau_j \left(R_{ij}^{-k} - l_{ik} f_{jk}\right)^2\right] \tag{47}$$

Expand the square $(R_{ij}^{-k} - l_{ik} f_{jk})^2$:

$$= \sum_{ij} \mathbb{E}_q\left[-\frac{1}{2}\tau_j \left(\cancel{\left(R_{ij}^{-k}\right)^2} - 2\, R_{ij}^{-k}\, l_{ik}\, f_{jk} + l_{ik}^2\, f_{jk}^2\right)\right] \tag{48}$$

The $(R_{ij}^{-k})^2$ term is constant w.r.t. f_{jk} and drops. Take the expectation $\mathbb{E}_{q_l}[\cdot]$ over L, treating R_{ij}^{-k} and τ_j as fixed:

$$= \sum_{ij} \left[\tau_j\, R_{ij}^{-k}\, \underbrace{\mathbb{E}_{q_l}[l_{ik}]}_{\bar{l}_{ik}}\, f_{jk} - \frac{1}{2}\tau_j\, \underbrace{\mathbb{E}_{q_l}[l_{ik}^2]}_{\overline{l_{ik}^2}}\, f_{jk}^2\right] \tag{49}$$

Now group over i (for each fixed j) to read off the quadratic form in f_{jk}:

$$\mathcal{F}(q_{f_k}) \;\approx\; \sum_j \left[-\frac{1}{2} f_{jk}^2 \underbrace{\left(\sum_i \tau_j\, \overline{l_{ik}^2}\right)}_{A_{jk}} + f_{jk} \underbrace{\left(\sum_i \tau_j\, R_{ij}^{-k}\, \bar{l}_{ik}\right)}_{B_{jk}}\right] + \mathbb{E}_{q_{f_k}}\left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)}\right] \tag{50}$$

6.3 Coefficients

Reading off from Eq. (50):

$$A_{jk} = \sum_i \tau_j\, \overline{l_{ik}^2}, \qquad B_{jk} = \sum_i \tau_j\, R_{ij}^{-k}\, \bar{l}_{ik} \tag{51}$$

6.4 EBNM Problem

Revision Note

[**R4**] Equations 47–48 of the 2/12/26 document subscripted the EBNM inputs as x_i, s_i^2 (sample index). Since we produce one pseudo-observation *per feature* when updating factor k (each j provides one equation), the correct subscript is j (feature index).

EBNM Problem

For each feature j, solve the EBNM problem:

$$x_j = \frac{B_{jk}}{A_{jk}} = \frac{\sum_i \tau_j \, R_{ij}^{-k} \, \bar{l}_{ik}}{\sum_i \tau_j \, \overline{l_{ik}^2}} \tag{52}$$

$$s_j = \frac{1}{\sqrt{A_{jk}}} \tag{53}$$

$$(\hat{g}_{f_k}, \, \hat{q}_{f_k}) \;=\; \mathrm{EBNM}(\mathbf{x}, \mathbf{s}) \tag{54}$$

7 Update for q_{β_k} — Survival Coefficients

7.1 Setting Up the Objective

$\boldsymbol{\beta}$ appears only in the survival term. Dropping the genomics log-likelihood (which is constant w.r.t. $\boldsymbol{\beta}$), the objective is:

$$\mathcal{F}(q_\beta, g_\beta) = \mathbb{E}_{q_l, q_\beta}[\log P(t, \delta \mid L, \boldsymbol{\beta})] + \sum_k \mathbb{E}_{q_{\beta_k}}\left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)}\right] \tag{55}$$

7.2 Preliminary: The Prior $g(\beta)$

The prior on $\boldsymbol{\beta}$ is a zero-mean normal:

$$\beta_k \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \qquad g(\beta_k) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\beta_k^2}{2\sigma^2}\right) \tag{56}$$

Taking the expectation of the log-prior:

Derivation Steps

$$\mathbb{E}_{q_\beta}[\log g(\boldsymbol{\beta})] = \mathbb{E}_{q_\beta}\left[\log \prod_k \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\beta_k^2}{2\sigma^2}\right)\right] \tag{57}$$

$$= \mathbb{E}_{q_\beta}\left[\sum_k \left(\underbrace{\log \frac{1}{\sqrt{2\pi\sigma^2}}}_{\text{constant, drops}} - \frac{\beta_k^2}{2\sigma^2}\right)\right] \tag{58}$$

$$= -\frac{1}{2\sigma^2}\,\mathbb{E}_{q_\beta}\left[\sum_k \beta_k^2\right] \tag{59}$$

7.3 Preliminary: The Variance-Mean Decomposition

A key identity used throughout is:

$$\mathbb{E}_q[l_{ik}^2] = \mathrm{Var}_q(l_{ik}) + (\mathbb{E}_q[l_{ik}])^2 = \mathrm{Var}_q(l_{ik}) + \bar{l}_{ik}^2 \tag{60}$$

which means $\overline{l_{ik}^2} = \mathrm{Var}_q(l_{ik}) + \bar{l}_{ik}^2 \geq \bar{l}_{ik}^2$. Using the posterior means as shorthand: $\mathbb{E}_q[l_{ik}] = \bar{l}_{ik}$ and $\mathbb{E}_q[l_{ik}^2] = \overline{l_{ik}^2}$.

7.4 Expanding the Survival Likelihood Over L

From the Taylor approximation with $\eta_i = \sum_k l_{ik}\beta_k$:

$$\mathcal{F}(q_\beta, g_\beta) = \mathbb{E}_{q_l,q_\beta}\left[\sum_i -\frac{1}{2}W_{ii}\left(z_i - \sum_k l_{ik}\beta_k\right)^2\right] + (\text{prior and posterior terms}) \tag{61}$$

Derivation Steps

Expand the square $(z_i - \sum_k l_{ik}\beta_k)^2$:

$$= \mathbb{E}_{q_l,q_\beta}\left[\sum_i -\frac{1}{2}W_{ii}\left(z_i^2 - 2z_i\sum_k l_{ik}\beta_k + \left(\sum_k l_{ik}\beta_k\right)^2\right)\right] \tag{62}$$

Expand the cross term. For any vector product:

$$\left(\sum_k l_{ik}\beta_k\right)^2 = \sum_k l_{ik}^2\beta_k^2 + 2\sum_{k<k'} l_{ik}\beta_k\, l_{ik'}\beta_{k'} \tag{63}$$

Revision Note

[R5] The 2/12/26 document (Eq. 62) wrote $(\sum_k l_{ik}\beta_k)^2 = \sum_k l_{ik}^2\beta_k + \ldots$, missing the exponent on β_k. The correct expansion is $\sum_k l_{ik}^2\beta_k^{\mathbf{2}} + 2\sum_{k<k'}\ldots$ as shown above.

Substituting back:

$$= \mathbb{E}_{q_l,q_\beta}\left[\sum_i -\frac{1}{2}W_{ii}\left(z_i^2 - 2z_i\sum_k l_{ik}\beta_k + \sum_k l_{ik}^2\beta_k^2 + 2\sum_{k<k'} l_{ik}\beta_k l_{ik'}\beta_{k'}\right)\right] \tag{64}$$

Now apply $\mathbb{E}_{q_l}[\cdot]$ (treat $\boldsymbol{\beta}$ as random under q_β, treat L as random under q_l). Under mean-field independence, $l_{ik} \perp l_{ik'}$ for $k \neq k'$, so:

$$\mathbb{E}_{q_l}[l_{ik}] = \bar{l}_{ik}, \quad \mathbb{E}_{q_l}[l_{ik}^2] = \overline{l_{ik}^2}, \quad \mathbb{E}_{q_l}[l_{ik}l_{ik'}] = \bar{l}_{ik}\bar{l}_{ik'} \text{ for } k \neq k' \tag{65}$$

Applying $\mathbb{E}_{q_l}[\cdot]$, the z_i^2 term drops (constant w.r.t. L):

$$\begin{aligned} &= \mathbb{E}_{q_\beta}\Bigg[\sum_i -\frac{1}{2}W_{ii}\Bigg(-2z_i\sum_k \bar{l}_{ik}\beta_k \\ &\qquad\qquad + \sum_k \overline{l_{ik}^2}\beta_k^2 + 2\sum_{k<k'} \bar{l}_{ik}\beta_k\bar{l}_{ik'}\beta_{k'}\Bigg)\Bigg] \end{aligned} \tag{66}$$

Recognise that the last two terms reconstitute the posterior-mean-based square:

$$\left(z_i - \sum_k \bar{l}_{ik}\beta_k\right)^2 = z_i^2 - 2z_i\sum_k \bar{l}_{ik}\beta_k + \sum_k \bar{l}_{ik}^2\beta_k^2 + 2\sum_{k<k'} \bar{l}_{ik}\bar{l}_{ik'}\beta_k\beta_{k'} \tag{67}$$

comparing with Eq. (66) we see an extra term:

$$\sum_k \overline{l_{ik}^2}\beta_k^2 - \sum_k \bar{l}_{ik}^2\beta_k^2 = \sum_k \mathrm{Var}_q(l_{ik})\,\beta_k^2 \tag{68}$$

Therefore (after dropping the constant z_i^2 term):

$$\begin{aligned} \mathbb{E}_{q_l,q_\beta}[\cdots] = \mathbb{E}_{q_\beta}\Bigg[&\sum_i -\frac{1}{2}W_{ii}\left(z_i - \sum_k \bar{l}_{ik}\beta_k\right)^2 \\ &+ \sum_i -\frac{1}{2}W_{ii}\sum_k \mathrm{Var}_q(l_{ik})\,\beta_k^2\Bigg] \end{aligned} \tag{69}$$

7.5 Combining All Terms

Substituting Eq. (59) and Eq. (69) into the full objective Eq. (55):

$$\mathcal{F}(q_\beta, g_\beta) = \mathbb{E}_{q_\beta}\left[\sum_i -\frac{1}{2}W_{ii}\left(z_i - \sum_k \bar{l}_{ik}\beta_k\right)^2\right] \tag{70}$$

$$- \mathbb{E}_{q_\beta}\left[\frac{1}{2}\sum_k\left(\sum_i W_{ii}\,\mathrm{Var}_q(l_{ik})\right)\beta_k^2\right] \tag{71}$$

$$- \frac{1}{2\sigma^2}\,\mathbb{E}_{q_\beta}\left[\sum_k \beta_k^2\right] \tag{72}$$

$$- \mathbb{E}_{q_\beta}[\log q(\boldsymbol{\beta})] \tag{73}$$

Derivation Steps

Combine terms (71) and (72) (both contain β_k^2):

$$-\frac{1}{2}\mathbb{E}_{q_\beta}\left[\sum_k\left(\sum_i W_{ii}\,\mathrm{Var}_q(l_{ik}) + \frac{1}{\sigma^2}\right)\beta_k^2\right] \tag{74}$$

The full objective is therefore:

$$\begin{aligned}\mathcal{F}(q_\beta, g_\beta) = \mathbb{E}_{q_\beta}&\left[\sum_i -\frac{1}{2}W_{ii}\left(z_i - \sum_k \bar{l}_{ik}\beta_k\right)^2\right]\\ &-\frac{1}{2}\mathbb{E}_{q_\beta}\left[\sum_k\left(\sum_i W_{ii}\,\mathrm{Var}_q(l_{ik}) + \frac{1}{\sigma^2}\right)\beta_k^2\right] - \mathbb{E}_{q_\beta}[\log q(\boldsymbol{\beta})]\end{aligned} \tag{75}$$

7.6 Coordinate-Ascent Update for β_k

Fix all $\beta_{k'}$ for $k' \neq k$. Define the survival partial working response (as in Eq. (29)):

$$z_i^{-k} = z_i - \sum_{k' \neq k} \bar{l}_{ik'}\bar{\beta}_{k'} \tag{76}$$

Derivation Steps

The first term in Eq. (75) contributes, for factor k:

$$\begin{aligned}
&\sum_i -\frac{1}{2} W_{ii} \left(z_i^{-k} - \bar{l}_{ik} \beta_k \right)^2 \\
&= \sum_i -\frac{1}{2} W_{ii} \left(\cancel{\left(z_i^{-k} \right)^2} - 2\, z_i^{-k}\, \bar{l}_{ik}\, \beta_k + \bar{l}_{ik}^2\, \beta_k^2 \right) \\
&\longrightarrow \underbrace{\left(\sum_i W_{ii}\, z_i^{-k}\, \bar{l}_{ik} \right)}_{B_k} \beta_k - \frac{1}{2} \underbrace{\left(\sum_i W_{ii}\, \bar{l}_{ik}^2 \right)}_{\text{part of } A_k} \beta_k^2
\end{aligned} \tag{77}$$

The variance-correction term (Eq. (74)) contributes, for factor k:

$$-\frac{1}{2} \left(\sum_i W_{ii}\, \mathrm{Var}_q(l_{ik}) + \frac{1}{\sigma^2} \right) \beta_k^2 \tag{78}$$

Combining (77) and (78) and using $\overline{l_{ik}^2} = \mathrm{Var}_q(l_{ik}) + \bar{l}_{ik}^2$:

$$\begin{aligned}
&-\frac{1}{2} \left(\sum_i W_{ii}\, \bar{l}_{ik}^2 + \sum_i W_{ii}\, \mathrm{Var}_q(l_{ik}) + \frac{1}{\sigma^2} \right) \beta_k^2 + \left(\sum_i W_{ii}\, z_i^{-k}\, \bar{l}_{ik} \right) \beta_k \\
&= -\frac{1}{2} \underbrace{\left(\sum_i W_{ii}\, \overline{l_{ik}^2} + \frac{1}{\sigma^2} \right)}_{A_k} \beta_k^2 + \underbrace{\left(\sum_i W_{ii}\, z_i^{-k}\, \bar{l}_{ik} \right)}_{B_k} \beta_k
\end{aligned} \tag{79}$$

Revision Note

[R6] The 2/12/26 document proposed a generic WLS solver for $\boldsymbol{\beta}$ without deriving the EBNM formulation. The coordinate-ascent derivation above (Eq. (79)) shows that each β_k update has the same quadratic EBNM structure as the L and F updates. Critically, A_k uses $\overline{l_{ik}^2}$ (the full second moment including posterior variance) rather than $\bar{l}_{ik}^2$ (the square of the posterior mean). This implements the *error-in-variables* correction: uncertainty in L inflates the effective noise for β_k, preventing over-fitting.

The combined precision and sufficient statistic are:

$$\boxed{A_k = \sum_i W_{ii}\, \overline{l_{ik}^2} + \frac{1}{\sigma^2}} \tag{80}$$

$$\boxed{B_k = \sum_i W_{ii}\, z_i^{-k}\, \bar{l}_{ik}} \tag{81}$$

EBNM Problem

For each factor k, solve the EBNM problem with a single pseudo-observation:

$$x_k = \frac{B_k}{A_k} = \frac{\sum_i W_{ii}\, z_i^{-k}\, \bar{l}_{ik}}{\sum_i W_{ii}\, \overline{l_{ik}^2} + 1/\sigma^2} \tag{82}$$

$$s_k = \frac{1}{\sqrt{A_k}} \tag{83}$$

$$(\hat{g}_{\beta_k},\, \hat{q}_{\beta_k}) \;=\; \mathrm{EBNM}(x_k,\, s_k) \tag{84}$$

Key remark. In the code, the $1/\sigma^2$ prior term is subsumed into the EBNM prior estimation — the EBNM solver fits g_{β_k} from data, so no fixed σ^2 is required. Using $\overline{l_{ik}^2}$ rather than $\bar{l}_{ik}^2$ in A_k encodes the error-in-variables correction.

7.7 Connection to Weighted Least Squares

The first term of Eq. (75) has the form of a weighted regression:

$$z_i = \sum_k \bar{l}_{ik}\, \beta_k + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}\big(0,\, W_{ii}^{-1}\big)$$

The coordinate-ascent EBNM update derived above is the natural Bayesian generalisation of this WLS step that: (i) accounts for posterior uncertainty in L via $\overline{l_{ik}^2}$, and (ii) applies an EBNM shrinkage prior on β_k.

8 Update for τ — Noise Precision

8.1 Objective

The components of the ELBO that depend on τ are:

Revision Note

[R7] Equation 83 of the 2/12/26 document wrote $\frac{1}{2}\sum_{ij}[\log \tau_{ij} + \tau_{ij}\bar{R}_{ij}^2]$. The sign on the second term is wrong: maximizing $\log\tau + \tau\bar{R}^2$ w.r.t. $\tau > 0$ gives no finite maximum (the function is unbounded above as $\tau \to \infty$). The correct expression has a **minus** sign.

$$\mathcal{F}(\tau) = \mathbb{E}_{q_l,q_f}\left[\sum_{ij}\left(\frac{1}{2}\log\tau_j - \frac{1}{2}\tau_j\left(Y_{ij} - \sum_k l_{ik} f_{jk}\right)^2\right)\right] + C = \frac{1}{2}\sum_{ij}\left[\log\tau_j - \tau_j\,\bar{R}_{ij}^2\right] + C \tag{85}$$

8.2 Expected Squared Residuals

Revision Note

[R8] Equation 85 of the 2/12/26 document wrote the last term as $\sum_k \overline{l^2_{jk}}\,\overline{f^2_{jk}}$, using the subscript jk on the loading. The loading carries a *sample* index i, not j.

The expected squared residual $\bar{R}^2_{ij}$ requires care because both L and F are random under q.

Derivation Steps

Starting from the definition:

$$\bar{R}^2_{ij} = \mathbb{E}_{q_l, q_f}\left[\left(Y_{ij} - \sum_k l_{ik} f_{jk}\right)^2\right] \tag{86}$$

Expand the square. Let $\mu_{ij} = \sum_k \bar{l}_{ik} \bar{f}_{jk}$ (the posterior mean reconstruction). Write $l_{ik} = \bar{l}_{ik} + \delta l_{ik}$ and $f_{jk} = \bar{f}_{jk} + \delta f_{jk}$ where $\mathbb{E}[\delta l_{ik}] = 0$ and $\mathbb{E}[\delta f_{jk}] = 0$.
Then:

$$Y_{ij} - \sum_k l_{ik} f_{jk} = Y_{ij} - \mu_{ij} - \sum_k \left(\bar{l}_{ik}\delta f_{jk} + \bar{f}_{jk}\delta l_{ik} + \delta l_{ik}\delta f_{jk}\right) \tag{87}$$

Taking the expectation of the square under mean-field ($l_{ik} \perp f_{jk'}$):

$$\bar{R}^2_{ij} = (Y_{ij} - \mu_{ij})^2 + \sum_k \bar{l}^2_{ik}\,\mathrm{Var}_q(f_{jk}) + \sum_k \bar{f}^2_{jk}\,\mathrm{Var}_q(l_{ik}) + \sum_k \mathrm{Var}_q(l_{ik})\,\mathrm{Var}_q(f_{jk}) \tag{88}$$

$$= \left(Y_{ij} - \sum_k \bar{l}_{ik}\bar{f}_{jk}\right)^2 + \sum_k \left[\overline{l^2_{ik}}\,\overline{f^2_{jk}} - \bar{l}^2_{ik}\,\bar{f}^2_{jk}\right] \tag{89}$$

The second step uses the identity $\overline{l^2_{ik} f^2_{jk}} - \bar{l}^2_{ik}\bar{f}^2_{jk} = \bar{l}^2_{ik}\mathrm{Var}_q(f_{jk}) + \bar{f}^2_{jk}\mathrm{Var}_q(l_{ik}) + \mathrm{Var}_q(l_{ik})\mathrm{Var}_q(f_{jk})$. The first term in Eq. (89) is the squared residual at posterior means; the second is the *variance inflation* correction.

$$\bar{R}^2_{ij} = \left(Y_{ij} - \sum_k \bar{l}_{ik}\bar{f}_{jk}\right)^2 + \sum_k \left[\overline{l^2_{\mathbf{ik}}}\,\overline{f^2_{jk}} - \bar{l}^2_{ik}\,\bar{f}^2_{jk}\right] \tag{90}$$

8.3 Deriving the Precision Estimator

Derivation Steps

Taking the derivative of $\mathcal{F}(\tau)$ in Eq. (85) with respect to τ_j and setting to zero:

$$\frac{\partial \mathcal{F}}{\partial \tau_j} = \frac{1}{2} \sum_i \left(\frac{1}{\tau_j} - \bar{R}_{ij}^2 \right) = 0 \tag{91}$$

Solving for τ_j:

$$\frac{n}{2\tau_j} = \frac{1}{2} \sum_i \bar{R}_{ij}^2 \quad \Rightarrow \quad \hat{\tau}_j = \frac{n}{\sum_i \bar{R}_{ij}^2} \tag{92}$$

8.4 Precision Estimators

$$\text{Column-specific } (\tau_{ij} = \tau_j): \quad \hat{\tau}_j = \frac{N}{\sum_i \bar{R}_{ij}^2} \tag{93}$$

$$\text{Constant } (\tau_{ij} = \tau): \quad \hat{\tau} = \frac{NP}{\sum_{ij} \bar{R}_{ij}^2} \tag{94}$$

In Eqs. (93) and (94):

- N = number of samples (rows of Y)
- P = number of features (columns of Y)
- The column-specific form estimates precision from N observations in column j
- The constant form averages across all NP entries

9 Complete Algorithm Summary

Supervised Matrix Factorization — CAVI Algorithm

Inputs: $Y_{n\times p}$, $(t_i, \delta_i)_{i=1}^n$, number of factors K
Initialize: $\bar{L}, \bar{F}$ via rank-K SVD of Y; $\bar{\boldsymbol{\beta}}$ via Cox on initial loadings; $\hat{\tau}$ via column-wise sample variance

For $b = 1, 2, \ldots$ **until convergence:**

1. Compute survival working quantities (one Taylor expansion per outer iteration):

$$\hat{\boldsymbol{\eta}} = \bar{L}\,\bar{\boldsymbol{\beta}}, \quad (u_i, W_{ii}) = \text{CoxTaylor}(\hat{\eta}_i, t_i, \delta_i), \quad z_i = \hat{\eta}_i + u_i/W_{ii}$$

2. For $k = 1, \ldots, K$:

(a) Update q_{l_k}, g_{l_k} (patient loadings):

$$\begin{aligned} R_{ij}^{-k} &= Y_{ij} - \textstyle\sum_{k'\neq k} \bar{l}_{ik'}\bar{f}_{jk'}, \quad z_i^{-k} = z_i - \textstyle\sum_{k'\neq k} \bar{l}_{ik'}\bar{\beta}_{k'} \\ A_{ik} &= \textstyle\sum_j \tau_j\,\overline{f_{jk}^2} + W_{ii}\,\overline{\beta_k^2}, \quad B_{ik} = \textstyle\sum_j \tau_j\,R_{ij}^{-k}\,\bar{f}_{jk} + W_{ii}\,z_i^{-k}\,\bar{\beta}_k \\ (\bar{l}_{ik}, \overline{l_{ik}^2}) &\leftarrow \text{EBNM}(x_i = B_{ik}/A_{ik},\ s_i = 1/\sqrt{A_{ik}}) \end{aligned}$$

(b) Update q_{f_k}, g_{f_k} (biological factors):

$$\begin{aligned} A_{jk} &= \textstyle\sum_i \tau_j\,\overline{l_{ik}^2}, \quad B_{jk} = \textstyle\sum_i \tau_j\,R_{ij}^{-k}\,\bar{l}_{ik} \\ (\bar{f}_{jk}, \overline{f_{jk}^2}) &\leftarrow \text{EBNM}(x_j = B_{jk}/A_{jk},\ s_j = 1/\sqrt{A_{jk}}) \end{aligned}$$

(c) Update q_{β_k}, g_{β_k} (survival coefficients):

$$\begin{aligned} z_i^{-k} &= z_i - \textstyle\sum_{k'\neq k} \bar{l}_{ik'}\bar{\beta}_{k'} \\ A_k &= \textstyle\sum_i W_{ii}\,\overline{l_{ik}^2}, \quad B_k = \textstyle\sum_i W_{ii}\,z_i^{-k}\,\bar{l}_{ik} \\ (\bar{\beta}_k, \overline{\beta_k^2}) &\leftarrow \text{EBNM}(x_k = B_k/A_k,\ s_k = 1/\sqrt{A_k}) \end{aligned}$$

3. Update $\hat{\tau}$ (column-specific noise precision):

$$\bar{R}_{ij}^2 = \big(Y_{ij} - \textstyle\sum_k \bar{l}_{ik}\bar{f}_{jk}\big)^2 + \textstyle\sum_k \left(\overline{l_{ik}^2}\,\overline{f_{jk}^2} - \bar{l}_{ik}^2\bar{f}_{jk}^2\right), \quad \hat{\tau}_j = \frac{N}{\sum_i \bar{R}_{ij}^2}$$

4. Check convergence:

$$\Delta_L = \max_k \|\bar{l}_k^{\text{new}} - \bar{l}_k^{\text{old}}\|_\infty < \text{tol}, \quad \Delta_\beta = \max_k |\bar{\beta}_k^{\text{new}} - \bar{\beta}_k^{\text{old}}| < \text{tol}$$

10 Errata Table — Summary of Revisions

ID	Location	Error in 2/12/26 doc	Correction
R1	Sec. 1A, model	Wrong dimension subscripts on Y, L, F, E	Eq. (3)
R2	Eq. 22	z_i^{-k} defined with $f_{jk'}$ instead of $\beta_{k'}$	Eq. (29)
R3	Eqs. 32–35	f_{jk}^2, β_k^2 should be posterior second moments	Eqs. (39)–(40)
R4	Eqs. 47–48	EBNM inputs subscripted i; should be j	Eqs. (52)–(53)
R5	Eq. 62	β_k missing square: should be β_k^2	Eq. (63)
R6	Sec. 6	β update as generic WLS; EBNM form missing	Eqs. (80)–(81)
R7	Eq. 83	Sign: $\log \tau + \tau \bar{R}^2$ should be $-\tau \bar{R}^2$	Eq. (85)
R8	Eq. 85	$\overline{l_{jk}^2}$ wrong index j; should be i	Eq. (90)

11 Notes on the Current R Implementation

Orthogonalization resets second moments. Every 10 iterations the code applies an SVD rotation to $\bar{L}$ for identifiability, then resets `EL2 <- EL^2` and `EF2 <- EF^2`. This drops the posterior variance, setting $\overline{l_{ik}^2} \leftarrow \bar{l}_{ik}^2$. The correct second moment is $\bar{l}_{ik}^2 + \mathrm{Var}_q(l_{ik}) \geq \bar{l}_{ik}^2$. The variance is recovered at the next EBNM call, so the algorithm remains consistent but may transiently underestimate posterior uncertainty every 10 iterations.

Taylor expansion computed once per outer iteration. The working quantities z_i and W_{ii} are computed once per outer loop before the inner k-loop, so they use the linear predictor $\hat{\boldsymbol{\eta}}$ from the *start* of each outer iteration. Within the inner loop, as $\bar{L}$ and $\bar{\boldsymbol{\beta}}$ are updated, the true linear predictor changes but z_i and W_{ii} are not refreshed. This is a standard approximation in IRLS-within-VI algorithms and is acceptable; the outer loop restores exactness at each iteration.

Single-observation EBNM for β_k. For each k, the β_k EBNM call uses a single pseudo-observation x_k. The EBNM package requires at least one observation, but the empirical prior g_{β_k} is estimated from just this one point, making the prior estimation highly variable. For larger K, this could be improved by pooling information across factors.